TÜBİTAK–2209-A ÜNİVERSİTE ÖĞRENCİLERİ ARAŞTIRMA PROJELERİ DESTEĞİ PROGRAMI

ARAŞTIRMA ÖNERİSİ FORMU

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1. Dönem Başvurusu

2209/A ÜNİVERSİTE ÖĞRENCİLERİ ARAŞTIRMA PROJELERİ DESTEĞİ PROGRAMI
ARAŞTIRMA ÖNERİSİ FORMU

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ÖZET

Türkçe özetin araştırma önerisinin (a) özgün değeri, (b) yöntemi, (c) yönetimi ve (d) yaygın etkisi hakkında bilgileri kapsaması beklenir. Her bir özet 450 kelime veya bir sayfa ile sınırlandırılmalıdır. Bu bölümün en son yazılması önerilir.

This project aims to develop a reinforcement learning (RL)-based decision support system that enables helicopters to fly safely and stably under changing environmental conditions. One of the major challenges faced by helicopters in real-world operations is the loss of control caused by sudden changes in wind direction and intensity, pressure differences, and increased vibrations—particularly in mountainous regions. In the literature, most flight control studies using RL methods have focused on fixed-wing unmanned aerial vehicles (UAVs) or simulation environments operating under ideal conditions. However, these studies have not sufficiently addressed the complex aerodynamic structure of helicopters or the effects of variable environmental conditions. This project seeks to fill that gap by creating a more realistic learning environment where environmental factors such as wind, pressure, and vibration are directly integrated into the model.

The originality of this project lies in its adaptation of RL algorithms to helicopter control while explicitly accounting for environmental variables. This allows the agent to learn not only to reach a target but also to perform safe maneuvers in response to environmental changes. As a result, the developed system will go beyond classical control approaches, enabling the helicopter to behave in a way that more closely reflects real-world flight conditions.

The project management plan is structured in stages. Each work package focuses on a specific task, with data collection, modeling, training, and evaluation processes designed to complement one another. The model dynamics will be calibrated using data obtained from the simulation environment, after which the RL agent will be trained under various scenarios. The performance of the resulting model will then be assessed through statistical analyses.

The potential impact of this project is broad. The developed model can be integrated into future autonomous flight safety systems for UAVs and helicopters. Such a system could contribute to preventing helicopter accidents, especially under challenging weather conditions. Furthermore, the simulation environment and RL model developed within this project will be shared as open-source tools for academic research and educational purposes. In this way, the project will not only enhance scientific knowledge but also guide future studies in the field of autonomous aerial systems.

Anahtar Kelimeler: Reinforcement Learning, Helicopter Control, Simulation, PPO, Autonomous Flight, Obstacle Avoidance, Artificial Intelligence

1. ÖZGÜN DEĞER

1.1. Konunun Önemi, Araştırma Önerisinin Özgün Değeri ve Araştırma Sorusu/Hipotezi

Araştırma önerisinde ele alınan konunun kapsamı ve sınırları ile önemi literatürün eleştirel bir değerlendirmesinin yanı sıra nitel veya nicel verilerle açıklanır.

Özgün değer yazılırken araştırma önerisinin bilimsel değeri, farklılığı ve yeniliği, hangi eksikliği nasıl gidereceği veya hangi soruna nasıl bir çözüm geliştireceği ve/veya ilgili bilim veya teknoloji alan(lar)ına kavramsal, kuramsal ve/veya metodolojik olarak ne gibi özgün katkılarda bulunacağı literatüre atıf yapılarak açıklanır.

Önerilen çalışmanın araştırma sorusu ve varsa hipotezi veya ele aldığı problem(ler)i açık bir şekilde ortaya konulur.

Subject

Aircraft with vertical takeoff and landing capabilities (helicopters and rotary-wing UAVs) are among the most critical elements of search and rescue, medical services, and military operations in mountainous and hard-to-reach areas. However, the most common causes of accidents in these missions include dangerous approaches to mountains and sudden wind/turbulence conditions. Wind changes on mountain slopes cause vibrations in the rotor system, disrupting the balance of the control surfaces and increasing the pilot's workload. This seriously limits the area in which the pilot can maneuver safely. In this context, the aircraft's ability to simultaneously analyze obstacles and weather conditions and select the safest and most appropriate maneuvers is seen as a fundamental goal that will ensure not only operational efficiency but also the protection of human life.

Gaps in the Literature and Problem Definition

Helicopters, especially single or twin-rotor ones, exhibit complex and highly dynamic behavior during flight. However, a significant portion of the literature on flight control and obstacle avoidance focuses on quadrotors (small quad-rotor UAVs) or fixed-wing UAVs. Since the dynamics of these vehicles are simpler compared to helicopters, the methods obtained do not sufficiently represent the actual flight characteristics of helicopters.

Current reinforcement learning (RL)-based studies have mostly been conducted under fixed or simplified environmental conditions. However, factors such as wind direction and intensity, rotor vibration, or pressure changes are constantly changing during actual flight; despite this, in many simulation environments, these variables are either idealized or completely ignored. This situation leads to significant limitations in the transfer of policies learned in the simulation environment to real flight (sim-to-real transfer).

Furthermore, studies on obstacle avoidance and maneuvering strategies specific to helicopters are quite limited. Existing research generally sticks to predefined scenarios and does not provide sufficient data on the agent's adaptation to variable environmental conditions. RL-based experimental applications for helicopters are also virtually non-existent; most existing examples have been conducted on quadrotors (e.g., Molchanov et al., 2019).

In summary, the literature indicates that the complex and unique flight dynamics of helicopters are not sufficiently modeled, and that adaptable maneuvering and obstacle avoidance strategies in response to environmental variables remain limited. Furthermore, significant difficulties are encountered in transferring policies learned in the simulation environment to real flight conditions, and it is noteworthy that RL approaches specifically designed for helicopters are quite inadequate. These gaps clearly demonstrate the scientific and practical contributions of this work. The proposed project aims to develop an RL agent for helicopters that is resilient to environmental variables, safety-focused, and capable of adaptive maneuvering.

Original Value (Scientific/Technological Contribution)

The original aspects that this study will contribute to the literature and practice can be summarized as follows:

- **Multi-parameter perception-decision framework:** The helicopter's distance from the mountain, wind conditions, and rotor vibrations will be monitored simultaneously and converted into a single risk indicator, which will provide input to the decision mechanism. Thus, the system will be able to evaluate different risk factors holistically, rather than separately.

- **Multi-objective reward design:** A reward function will be developed that simultaneously considers time, route length, and fuel consumption without compromising safety. This function will penalize safe distance violations while also balancing efficiency targets.

- **3D simulation and visualization:** In a PyPlot-based simulation environment, the flight path, wind vectors, and selected maneuvers will be tracked in real time with three-dimensional visuals. This will allow the performance of the developed approach to be presented in a concrete and visual manner.

- **Phased architecture:**

    1. Continuous measurement of distance to the mountain and warning based on adaptive thresholds,

2. Selection of appropriate maneuvers considering environmental parameters such as wind and vibration,

3. Route and energy optimization aimed at reaching the safest destination in the shortest time with the least fuel.

Project Questions: Can an autonomous and preventive decision-making mechanism be designed to counter the risk of mountain collisions, which cause the vast majority of helicopter crashes, unlike existing rule-based warning systems? Can this decision-making mechanism, using artificial intelligence-based Reinforcement Learning (RL) methods, learn the safest and most efficient avoidance maneuver even under environmental uncertainties (wind direction, wind speed, pressure, etc.)? Can this method provide superiority in terms of safety and efficiency compared to rule-based classical approaches?

Hypothesis: When trained using the FlightGear simulation environment, utilizing the helicopter's instantaneous dynamic state and terrain geometry, algorithms based on Reinforcement Learning (RL) principles will learn a superior, robust, and reliable autonomous policy compared to traditional rule-based avoidance systems. This trained RL-based decision-making mechanism will maximize collision-free flight safety and offer a more efficient (less fuel consumption), safer (fewer obstacle violations), and more economical (shorter time and distance) flight than rule-based methods, as environmental factors such as wind direction, wind speed, and pressure parameters are considered simultaneously. more efficient (shorter time and distance) compared to rule-based methods.

1.2. Amaç ve Hedefler

Araştırma önerisinin amacı ve hedefleri açık, ölçülebilir, gerçekçi ve araştırma süresince ulaşılabilir nitelikte olacak şekilde yazılır.

Project Questions: Can an autonomous and preventive decision-making mechanism be designed to counter the risk of colliding with mountains, which causes the vast majority of helicopter crashes, unlike existing rule-based warning systems? Can this decision-making mechanism, using Artificial Intelligence-based Reinforcement Learning (RL) methods, learn the safest and most efficient avoidance maneuver even under environmental uncertainties (wind direction, wind speed, pressure, etc.)? Can this method provide superiority in terms of safety and efficiency compared to rule-based classical approaches?

Hypothesis: Algorithms based on Reinforcement Learning (RL) principles, when trained using the helicopter's instantaneous dynamic state and terrain geometry in the FlightGear simulation environment, will learn a superior, robust, and reliable autonomous policy compared to traditional rule-based avoidance systems. This trained RL-based decision-making mechanism will maximize collision-free flight safety while simultaneously considering environmental factors such as wind direction, wind speed, and pressure parameters. It will also be more efficient (less fuel consumption), safer (fewer obstacle violations), more efficient (shorter time and distance) compared to rule-based methods.

Project Objective:

General objective: To directly contribute to solving the problem of mountain collisions, one of the most significant and critical causes of helicopter crashes. The fundamental goal is to develop an autonomous decision support and avoidance system that combines engineering and artificial intelligence disciplines in a concrete and verifiable approach, going beyond traditional warning systems to address this ongoing vital threat to aviation safety worldwide. The RL-based system to be developed will not only prevent collisions but also aim to reduce pilot workload by learning the most appropriate and robust maneuvers even under environmental uncertainties (wind direction, wind speed, pressure, etc.). Ultimately, this project aims to offer a groundbreaking technological solution in aviation safety, serving as a pioneering study on the integration of autonomous systems into safety-critical aircraft.

Specific objective: To design an autonomous control system that determines the most appropriate maneuver based on data from environmental and structural parameters and optimizes the flight route

according to multi-faceted criteria in order to ensure the safe navigation of the helicopter in mountainous terrain. Within this framework, the helicopter's collision avoidance capability will be demonstrated in a simulation environment using Reinforcement Learning methods. The goal is to create a robust, adaptive, and reliable flight control system that maintains high performance even under external conditions such as wind and rotor vibration, using these developed autonomous maneuvering strategies. In this way, the project will lay the groundwork for the development of intelligent and safe autonomous control systems to be used in the aviation industry in the future.

Project Goals:

- To create a distance detection module that continuously measures the helicopter's distance from the mountain and provides a warning when critical limits are reached.
- To incorporate parameters such as wind direction, wind speed, and pressure into the helicopter's maneuver decision-making mechanism.
- Determine the helicopter's directional maneuvers using an RL-based decision mechanism.
- Define a multi-objective reward function that optimizes safety, fuel consumption, and flight duration simultaneously.
- Create a three-dimensional simulation environment using the FlightGear simulator to enable visual analysis of flights.
- Demonstrate the generalizability of the developed model by testing its applicability under different environmental conditions.

Main Supporting Arguments (Justifications):

- Different wind profiles and mountainous terrain conditions increase the helicopter's safety risks..
- Vibration parameters and rotor behavior play a critical role in the decision-making mechanism.
- RL-based learning can yield better results than rule-based methods in uncertain and variable conditions.
- RL-based learning can yield better results than rule-based methods in uncertain and variable conditions.
- The model tested in a simulation environment can provide valuable safety insights prior to actual flights.
- Findings that contribute to aviation safety and accident prevention may pave the way for more extensive studies in the future.

2. **YÖNTEM**

Araştırma önerisinde uygulanacak yöntem ve araştırma teknikleri (veri toplama araçları ve analiz yöntemleri dahil) ilgili literatüre atıf yapılarak açıklanır. Yöntem ve tekniklerin çalışmada öngörülen amaç ve hedeflere ulaşmaya elverişli olduğu ortaya konulur.

Yöntem bölümünün araştırmanın tasarımını, bağımlı ve bağımsız değişkenleri ve istatistiksel yöntemleri kapsaması gerekir. Araştırma önerisinde herhangi bir ön çalışma veya fizibilite yapıldıysa bunların sunulması beklenir. Araştırma önerisinde sunulan yöntemlerin iş paketleri ile ilişkilendirilmesi gerekir.

The proposed research methodology is based on the development of a reinforcement learning (RL)-based decision support system and its testing in a 3D simulation environment. The study will involve both theoretical modeling and simulation-based validation. The methodology encompasses the definition of dependent and independent variables, data collection and analysis procedures, research steps divided into work packages, and statistical evaluations.

Independent Variables: Wind direction and speed, obstacle positions and movement states, target position, helicopter current position and orientation (x, y, z, roll, pitch, yaw), IMU data (acceleration, angular velocity, vibration energy), motor output limits, sensor noise levels.

Dependent Variables: Helicopter maneuvering stability, number of collisions, target reach success rate, energy consumption (fuel/power efficiency), mission completion time, mean reward per episode, fuel/energy usage.

Tools:

- **Simulation Environment:** AirSim or Gazebo (to provide realistic physics-based simulation)
- **RL Framework:** Stable Baseline3 (for PPO, SAC, DDPG algorithms)
- **Programming Language:** Python
- **Data Analysis:** Pandas, NumPy, Matplotlib
- **Statistical Evaluation:** SciPy, scikit-learn (ANOVA, t-test, correlation)
- **Visualization:** Matplotlib, Seaborn
- **Model Monitoring:** TensorBoard (for learning curves and reward trend analysis)

Statistical Analysis:

- **Success Rate Analysis:** The percentage of target reach during training will be measured.
- **Reward Curve Analysis:** Mean episode reward values will be plotted over time. An increasing trend indicates successful learning. Energy consumption of different algorithms for the same task will be compared, and a one-way ANOVA test will be applied to determine whether differences between algorithms are statistically significant.
- **Stability Measurement:** The standard deviation of roll and pitch angles during flight will be calculated. Lower deviation indicates more stable maneuvers.
- **Collision Analysis:** Correlation analysis will be performed between the number of collisions per episode and reward drops.

Material Details:

- **Programming Language:** Python
- **Libraries:** NumPy, PyTorch/TensorFlow (for RL), Matplotlib (3D PyPlot), Gym environment (customized)
- **RL Algorithms:** Deep Q-Network (DQN), Proximal Policy Optimization (PPO)
- **Hardware:** GPU-enabled computer (for training)

Methodology Details:

The aim of this project is to address the helicopter-specific obstacle avoidance problem and maneuverability under dynamic environmental conditions (such as wind, acoustic vibrations, pressure changes, etc.) using a

Reinforcement Learning (RL) agent. There are several studies in the literature applying RL to helicopters; one notable example is the work by Abbeel et al., where aerobatic maneuvers were successfully performed on a real helicopter, demonstrating that helicopter dynamics are suitable for learning ["An Application of Reinforcement Learning to Aerobatic Helicopter Flight" (Abbeel et al., NIPS 2007)]. Therefore, RL has been chosen as the core approach for this project.

Project Methodology and Work Packages:

1. **Literature Review and Problem Definition**
2. **Data Collection and System Identification using Simulation Tools**
3. **RL Environment and Agent Design**
4. **Algorithm Selection and Agent Training**
5. **Training Process and Applied Approaches**
6. **Domain Randomization – Preparation for Real-World Deployment**
7. **Safe Training and Uncertainty Management**
8. **Expected Results and Evaluation**

These stages have been defined and described in detail to provide a structured framework for the project.

Work Package 1 - Literature Review and Problem Definition

In this work package, approximately 20 academic studies on helicopter control, obstacle avoidance, RL algorithms (PPO, DDPG, SAC), and simulation-to-reality transfer (sim-to-real) methods were reviewed. The literature revealed that rule-based approaches are insufficient in real flights.

As a result of this analysis, it was concluded that an RL-based control system should be developed to enable helicopters to maneuver safely, stably, and efficiently under variable environmental factors (wind, vibration, sensor noise, pressure).

The findings from this work package will form the basis for the methods to be used in subsequent stages.

Work Package 2 - Data Collection Using Simulation Tools and System Identification

Due to the limited availability of real flight data, training cannot be conducted with actual helicopter flights. Therefore, data will be obtained using physics-based simulators such as **AirSim** or **FlightGear**. These simulators allow modeling of helicopter rotor dynamics, sensor models, and environmental factors to generate flight data. The collected data, including acceleration, orientation (roll, pitch, yaw), rotor speed, and wind pressure, will be used to calibrate the physical parameters within the simulation.

In real helicopter flights, wind conditions, vibration levels, and atmospheric pressure are never constant. However, in simulation, these parameters are often considered ideal, which allows the agent to learn easily in the simulated environment but may lead to failure in real-world conditions. To mitigate this, the variations in wind direction and intensity, rotor vibrations, and pressure fluctuations are statistically modeled based on the collected flight data. These models are then incorporated into the training environment, ensuring that the agent experiences different environmental conditions in each trial. As a result, the agent learns to reach the target not only in a single scenario but across multiple scenarios, achieving this with a more robust and flexible control policy. Similar approaches in the literature have demonstrated that adjusting physical variables using simulation data plays a key role in reducing the sim-to-real gap [*Sim-to-(Multi)-Real: Transfer of Low-Level Robust Control Policies to Multiple Quadrotors, Artem Molchanov et al.*].

Implementation Steps (Data Collection):

1. Multiple flight scenarios will be executed under varying payloads, wind conditions, and rotor parameters. Flight data will be recorded in CSV/ULog format, including position, velocity, angular velocity, timestamp, etc. These data will reveal how the helicopter behaves under different flight conditions.

2. The collected data cannot be used directly in the first stage due to potential sensor errors or missing records. Therefore, the data will be cleaned, synchronized, and formatted appropriately.

3. The processed data will then be used to improve the accuracy of the helicopter's physical model. Specifically, values such as weight distribution, rotational inertia, and motor responses in the simulation will be adjusted according to the flight data. This ensures that the agent flying in the simulation behaves much more like a real helicopter.

Work Package 3 - RL Environment and Agent Design

In the RL environment we will create, the agent will perceive the current state of the helicopter, generate appropriate commands, and gradually improve its flight strategy through a reward mechanism.

State (Observation) Design:
To enable the agent to understand the helicopter's current state, an observation vector will be constructed. This vector will include both the helicopter's flight dynamics and the environmental conditions. The observation set will consist of:

- Position differences between the agent and the target (dx, dy, dz)
- Linear velocities (vx, vy, vz)
- Angular velocities (p, q, r)
- Euler angles (roll, pitch, yaw)
- Vibration measurements obtained from IMU data
- Wind predictions from the simulation

This observation set will allow the agent not only to navigate toward the target but also to maintain flight safety and avoid collisions with terrain or obstacles.

Action Design:

During the initial stages of learning, the agent will use high-level control commands instead of sending signals directly to the motors. In other words, rather than calculating individual PWM signals for each rotor, the agent will work with abstract commands such as target angular velocity or target speed. This approach simplifies the learning process. By interacting with the environment through direct movement and speed commands, the agent can learn faster and more efficiently, while maintaining stability. The risk of the helicopter falling or losing balance is significantly reduced in the early stages, providing a safer training environment. Additionally, this approach allows for gradual refinement, enabling the agent to eventually transition to lower-level motor control as training progresses. This method has been shown to be effective in the literature; for example, in quadrotor studies, agents first learned basic flight skills through high-level commands and later refined precise maneuvers at the motor level [*Sim-to-(Multi)-Real: Transfer of Low-Level Robust Control Policies to Multiple Quadrotors, Artem Molchanov et al.*].

Reward and Penalty Design:
The reward system for the agent will be designed to encourage the helicopter to reach the target both safely and efficiently. Rewards will vary depending on different flight scenarios:

- **Approaching the target:** The helicopter receives positive rewards as it gets closer to the target, teaching the agent to navigate toward the goal.
- **Collision with terrain or obstacles:** Large penalties (negative rewards) will be applied if the helicopter collides with mountains or other obstacles, encouraging collision avoidance.
- **Correct maneuvers against wind:** Small to moderate positive rewards will be given if the helicopter performs maneuvers correctly in response to environmental factors like wind. This encourages the agent to act intelligently under dynamic conditions.

- **Reaching the target:** A large positive reward will be granted upon successfully reaching the target, reinforcing the ultimate objective of the learning process.

This multi-component reward design ensures that the agent learns not only to reach the target but also to consider flight safety, efficiency, and environmental adaptation during navigation. The contribution of each behavior to the total reward will be fine-tuned through trial-and-error and hyperparameter search methods such as grid search.

Work Package 4 - Algorithm Selection

In this project, **Proximal Policy Optimization (PPO)** has been primarily selected for teaching the helicopter maneuvering and obstacle avoidance tasks. PPO provides stable and reliable learning in systems requiring continuous control and is relatively straightforward to implement (Schulman et al., 2017). In scenarios where the agent must control continuously varying variables such as the helicopter's velocity, orientation, and angular movements, PPO helps the agent learn accurately and consistently. In the literature, PPO has repeatedly demonstrated success in robotic manipulation and flight control applications in terms of stability and efficiency. Therefore, PPO is a logical first choice for this project.

Alternatively, **Deep Deterministic Policy Gradient (DDPG)** and Soft Actor-Critic (SAC) algorithms will also be considered. DDPG is a pioneering method for continuous control problems and is particularly advantageous in situations requiring precise motor-level control (Lillicrap et al., 2015). SAC, on the other hand, enriches learning with an entropy-based reward mechanism, enabling the agent to learn more stably and reliably.

During the project, these algorithms will be applied in a progressive manner: initially, PPO will be used to teach the helicopter basic flight and target-following policies; subsequently, DDPG or SAC will be applied for fine-tuning more precise maneuvers and low-level motor commands. This stepwise approach ensures that the agent first learns a generally stable policy and later achieves high performance in complex and precise tasks.

This methodology is commonly employed in the literature for successfully transferring agents trained in simulation to real-world conditions and is preferred to ensure a safe and effective learning process (Schulman et al., 2017; Lillicrap et al., 2015).

Work Package 5 - Training Process and Applied Approaches

The AI model developed in this project will be managed gradually as an RL-based system. That is, the model will not be exposed to complex scenarios initially; instead, it will first learn basic behaviors in simple environments and progressively improve in increasingly complex scenarios. This approach is known in the literature as curriculum learning, and multiple studies have shown that it accelerates learning and enhances the model's generalization capability [Bengio et al., 2009; Narvekar et al., 2020].

Stage 1 – Simple Scenario (Basic Flight Behaviors):
In the first stage, the environment will be kept very simple, and the agent will only learn to navigate toward the target and avoid a static obstacle. This allows the model to safely acquire fundamental behaviors, such as reaching the target, maintaining stability, and performing basic maneuvers. Wind and sensor information will be kept at minimal levels during this stage.

Stage 2 – Medium Difficulty Scenario (Generalization Phase):
At this stage, environmental conditions will be diversified by introducing different obstacle configurations, mild wind currents, and sensor data. This ensures that the agent learns to make correct decisions not only under fixed conditions but also in dynamic environments.

Stage 3 – Hard Scenarios (Realistic Conditions):
In the final stage, the environment will be designed to closely resemble real-world conditions. Dynamic obstacles, sudden wind changes, sensor drifts, and vibrations will be incorporated. This enables the agent to learn stable flight even in complex and realistic scenarios.

Work Package 6 - Domain Randomization – Preparation for Real-World Deployment

The agent will be trained not only in predefined scenarios but also under a variety of conditions. For this purpose, **domain randomization** will be employed. This technique randomly varies certain variables in the simulation environment during each training episode, exposing the agent to a different environment every time. The goal is to ensure that the model performs equally well in real-world flights. Models trained in a fixed

simulation environment often fail to adapt to the variable conditions encountered during actual flights. Domain randomization has been found to be highly effective in mitigating this sim-to-real transfer problem (Tobin et al., 2017; Sadeghi & Levine, 2017).

Work Package 7 - Safe Training and Uncertainty Management

Since mistakes during flight can be costly, the training process will be designed to be safe and cautious. The model will evaluate the uncertainty associated with each action and learn to avoid high-risk maneuvers. For example, if the model approaches a mountain too closely or encounters inconsistencies in sensor readings, the situation will be marked as “high risk,” and the system will penalize such actions. This approach is known in the literature as uncertainty-aware reinforcement learning and has been shown to improve safety in robotic systems (Berkenkamp et al., 2017). Consequently, the model will exhibit more cautious, safe, and stable flight behavior before transitioning to real hardware tests.

Work Package 8 - Expected Results and Evaluation

The primary goal of this project is to ensure that the helicopter reaches its target safely, stably, and efficiently under varying wind conditions and obstacle configurations. The Proximal Policy Optimization (PPO) algorithm, which will be employed in this study, is expected to provide the most balanced results in terms of maneuver stability and reward optimization. Throughout the training process, the agent will reinforce correct maneuvers and reduce erroneous behaviors based on feedback from the environment (reward-penalty system). For instance, the helicopter will receive positive rewards when it executes proper directional movements toward the target or maintains balance against wind forces, while collisions with obstacles, deviation from the planned trajectory, or unstable maneuvers will incur negative rewards, discouraging the repetition of such actions.

The model trained in the simulation environment is intended to generalize to conditions similar to real flights. To achieve this, the domain randomization approach will be applied, continuously varying parameters such as wind speed, sensor noise, and obstacle placement. This ensures that the agent becomes more robust and can perform successfully not only in a single environment but also under diverse physical conditions.

As training progresses, it is expected that the helicopter’s average reward will increase, the collision rate will decrease, and energy consumption will be optimized. The model’s performance will be evaluated using multiple metrics, including target reach success rate, mission completion time, energy efficiency, collision frequency, and adaptability to wind conditions. Each metric will be regularly monitored during both training and testing phases, and the results will be compared across different algorithms (e.g., PPO, DDPG, SAC).

In conclusion, the anticipated outputs are an autonomous helicopter model capable of stable maneuvers, resilient to environmental variations, and exhibiting safe flight behavior. The algorithm demonstrating the best performance will be selected as the core control policy for sim-to-real transition, providing a strong foundation for both academic research and practical implementation.

3 PROJE YÖNETİMİ

3.1 İş- Zaman Çizelgesi

Araştırma önerisinde yer alacak başlıca iş paketleri ve hedefleri, her bir iş paketinin hangi sürede gerçekleştirileceği, başarı ölçütü ve araştırmanın başarısına katkısı "İş-Zaman Çizelgesi" doldurularak verilir. Literatür taraması, gelişme ve sonuç raporu hazırlama aşamaları, araştırma sonuçlarının paylaşımı, makale yazımı ve malzeme alımı ayrı birer iş paketi olarak gösterilmemelidir.
Başarı ölçütü olarak her bir iş paketinin hangi kriterleri sağladığında başarılı sayılacağı açıklanır. Başarı ölçütü, ölçülebilir ve izlenebilir nitelikte olacak şekilde nicel veya nitel ölçütlerle (ifade, sayı, yüzde, vb.) belirtilir.

İŞ-ZAMAN ÇİZELGESİ (*)

İP No	İş Paketlerinin Adı ve Hedefleri	Kim(ler) Tarafından Gerçekleştirileceği	Zaman Aralığı (..-.. Ay)	Başarı Ölçütü ve Projenin Başarısına Katkısı
1	**Literature Review and Problem Definition**	Emine, Hatice, Merve	Week 1–2	At least 20 academic resources were reviewed, a literature summary and problem definition report were prepared.
2	**Data collection and system diagnostics using simulation tools**	Emine, Hatice, Merve	Week 2–4	By running simulation scenarios, a data set was created and system dynamics were extracted, providing a realistic infrastructure for the RL environment.
3	**RL Environment and Agent Design**	Emine, Hatice, Merve	Week 4–8	The RL environment has been established and the agent has been defined sensitive to environmental parameters and the decision-making infrastructure has been developed.
4	**Algorithm Selection and Agent Training**	Emine, Hatice, Merve	Week 8–10	Two different RL algorithms were implemented and trained, and the most appropriate method was determined by comparing performance.
5	**Training Process and Applied Approaches**	Emine, Hatice, Merve	Week 10–12	By testing different reward functions, the approach that provides the best results in safety, time and fuel optimization was selected.
6	**Domain Randomization – Preparation for Real-World Deployment**	Emine, Hatice, Merve	Week 12–14	Variables such as wind, turbulence and vibration were randomly integrated into the training environment, increasing the generalizability of the model.

7	**Safe Training and Uncertainty Management**	Emine, Hatice, Merve	Week 14–16	Security constraints were added to the RL training, ensuring that the model makes safe decisions under uncertainty conditions.
8	**Expected Results and Evaluation**	Emine, Hatice, Merve	Week 16–18	Simulation results were analyzed and compared with classical methods, and the unique contribution of the project was concretized.

(*) Çizelgedeki satırlar ve sütunlar gerektiği kadar genişletilebilir ve çoğaltılabilir.

3.2 Risk Yönetimi

Araştırmanın başarısını olumsuz yönde etkileyebilecek riskler ve bu risklerle karşılaşıldığında araştırmanın başarıyla yürütülmesini sağlamak için alınacak tedbirler (B Planı) ilgili iş paketleri belirtilerek ana hatlarıyla aşağıdaki Risk Yönetimi Tablosu'nda ifade edilir. B planlarının uygulanması araştırmanın temel hedeflerinden sapmaya yol açmamalıdır.

RİSK YÖNETİMİ TABLOSU*

İP No	En Önemli Riskler	Risk Yönetimi (B Planı)
1	The simulation environment (3D PyPlot and RL integration) was more complex and time consuming than expected.	Starting with a simpler 2D simulation environment, basic decision mechanisms will be tested there first, and then the transition to the 3D model will be made.
2	The RL algorithm (DQN/PPO) does not perform as expected or the training time is too long.	Alternatively, simpler rule-based methods and classical algorithms (A*, potential field) will be applied; comparison with RL will continue.
3	Accuracy issues in modeling environmental parameters (wind, vibration).	Standard wind models in the literature will be used and analysis will be done on simplified parameters.
4	RL training takes a long time due to hardware (GPU/processor) insufficiency.	A less complex network architecture will be chosen, training scenarios will be reduced and simplified, and cloud-based free GPU services (Colab, Kaggle) will be used.
5	Time management: Failure to complete work packages within the planned time.	Work packages will be executed in parallel, with flexible task distribution among team members (Emine, Hatice, Merve). Priority will be given to critical work packages (1–3).

(*) Tablodaki satırlar gerektiği kadar genişletilebilir ve çoğaltılabilir.

3.3. Araştırma Olanakları

Bu bölümde projenin yürütüleceği kurum ve kuruluşlarda var olan ve projede kullanılacak olan altyapı/ekipman (laboratuvar, araç, makine-teçhizat, vb.) olanakları belirtilir.

ARAŞTIRMA OLANAKLARI TABLOSU (*)

Kuruluşta Bulunan Altyapı/Ekipman Türü, Modeli (Laboratuvar, Araç, Makine-Teçhizat, vb.)	**Projede Kullanım Amacı**
Computer Laboratory (high-capacity desktop computers)	Training RL algorithms, running simulations
GPU-supported computers (NVIDIA CUDA compatible graphics cards)	Faster training of deep reinforcement learning algorithms
Python and related libraries (NumPy, PyTorch/TensorFlow, Matplotlib, Gym)	Simulation development, implementation of RL algorithms, visualization

University internet and software access (additional resources such as Google Colab, Kaggle)	In addition to GPU infrastructure, accelerating the training process, cloud-based experiments
Existing software/licenses of the project implementing institution (e.g. Flight Gear or similar)	Supporting use for alternative simulation and analysis studies
Study room and meeting infrastructure	For teamwork, reporting and project coordination

(*) Tablodaki satırlar gerektiği kadar genişletilebilir ve çoğaltılabilir.

4. **YAYGIN ETKİ**

Önerilen çalışma başarıyla gerçekleştirildiği takdirde araştırmadan elde edilmesi öngörülen ve beklenen yaygın etkilerin neler olabileceği, diğer bir ifadeyle yapılan araştırmadan ne gibi çıktı, sonuç ve etkilerin elde edileceği aşağıdaki tabloda verilir.

ARAŞTIRMA ÖNERİSİNDEN BEKLENEN YAYGIN ETKİ TABLOSU

Yaygın Etki Türleri	Önerilen Araştırmadan Beklenen Çıktı, Sonuç ve Etkiler
Scientific/Academic (Article, Proceedings, Book Chapter, Book)	At least one scientific article will be prepared as a result of the project and published in national/international conferences or TR Index journals. Project results will also be presented as posters or oral presentations at academic events.
Economic/Commercial/Social (Product, Prototype, Patent, Utility Model, Production Permit, Variety Registration, Spin-off/Start-up Company, Audiovisual Archive, Inventory/Database/Documentation Production, Copyrighted Work, Media Appearance, Fair, Project Market, Workshop, Training, etc. Scientific Event, Institution/Organization that Will Use Project Results, etc. Other common impacts)	Awareness will be raised on simulation-based decision support systems to prevent helicopter accidents. The developed model will form a foundational infrastructure that can be used in future aviation training and helicopter safety systems.
Researcher Training and Creating New Project(s) (Master's/Doctoral Thesis, National/International New Project)	This project will enhance the research skills of team members, providing them with the knowledge and experience that will form the basis for future master's and doctoral projects. It will also serve as a pioneer for more comprehensive aviation security projects.

5. BÜTÇE TALEP ÇİZELGESİ

Bütçe Türü	Talep Edilen Bütçe Miktarı (TL)	Talep Gerekçesi
Sarf Malzeme	4000	Materials required for the project but missing from the Department's laboratory have been requested. Other necessary materials will be used from the department's inventory.
Makina/Teçhizat (Demirbaş)		
Hizmet Alımı		
Ulaşım		
TOPLAM		

NOT: Bütçe talebiniz olması halinde hem bu tablonun hem de TÜBİTAK Yönetim Bilgi Sistemi (TYBS) başvuru ekranında karşınıza gelecek olan bütçe alanlarının doldurulması gerekmektedir. Yukardaki tabloda girilen bütçe kalemlerindeki rakamlar ile, TYBS başvuru ekranındaki rakamlar arasında farklılık olması halinde TYBS ekranındaki veriler dikkate alınır ve başvuru sonrasında değiştirilemez.

6. BELİRTMEK İSTEDİĞİNİZ DİĞER KONULAR

Sadece araştırma önerisinin değerlendirilmesine katkı sağlayabilecek bilgi/veri (grafik, tablo, vb.) eklenebilir.

7. EKLER

EK-1: KAYNAKLAR